Monitoring

English version of ../../operations/monitoring.md.

Signal	Where	Purpose
`GET /`	application	Render health-check (`healthCheckPath: /`)
stdout logs	process	route errors, keep-alive messages
`quiz_submissions` table	SQLite	request flow and statuses
keep-alive log	stdout, when enabled	diagnosing sleep on the free tier

Minimal alerts

1. `/` does not answer for 2 minutes (Render restarts the service by itself).
2. The 5xx share is above 1% over 5 minutes.
3. SMTP errors while credentials are configured.
4. A growing number of 409 responses on sign-up — brute force or a client bug.

What is missing

- No `/metrics`, no tracing, no SLO.
- No structured logs: free-form printing to stdout.
- No notification about new requests apart from e-mail (if configured).